

Applause Meter Report

ECE110

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Abstract

An applause meter provides a simple, effective method of converting sound intensity into a visual presentation through LEDs. This project reflects the concepts learned throughout the ECE 110 Lab, as we utilize concepts to build, test, and debug circuits. In this project our team built and tested circuitry that we deemed to be fit. Through equipment like oscilloscopes, we help test and verify each sub-circuit to show proof of the overall performance of each of the sub-circuits.

Introduction

Sound-level indicators are necessary in commonly used audio equipment, scientific equipment, and even common consumer electronics we may not consider needing it. These applause meters can come in a variety of ways. Some are designed as IC circuits like the LM3914 by Texas instrument that uses LEDs to visually represent the volume. [1]

In this project, the applause meter is created through analog circuits with the components we learned throughout the ECE 110 Lab, such as op-amp amplification and the utilization of MOSFETs. This report presents the theoretical basis that allows the creation of such a project in the first place, the implementation, and experimental verification of each sub-circuit, as well as a discussion of the challenges when integrating all the sub-circuits into one functioning circuit.

Module 1:

I worked on the microphone and Amplifier for the applause meter. The microphone converts sound into an electric signal where it then can be measured for our circuit. It can do this by having the sound waves vibrate a diaphragm which can create an induced current. This current can also be seen as a voltage which is our signal. It could also be seen as a capacitor in a way. When the front plate vibrates, it changes the distance between the electret material and the backplate. This changes the capacitance of the material which creates an induced voltage.

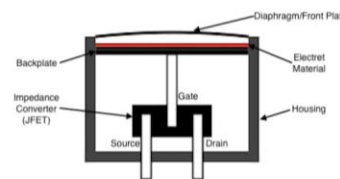


Figure 1.1 Cross section of a Condenser Microphone [2]

Here was the IV curve that I was able to create to try and map out the Thevenin equivalence when using testing points of resistance of the circuit and the resulting voltage. I used a 9-volt power supply with varying resistance of: $10\text{k}\ \Omega$, $1\text{k}\ \Omega$, and $2.2\text{k}\ \Omega$. From the graph, it appears that the slope of the IV was around -1 with a short circuit current of around 0.01 amps.

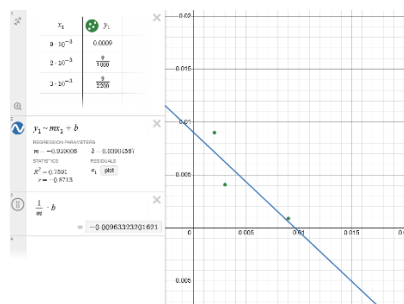


Figure 1.3 IV Characteristic curve found by using line of best fit on test point



Figure 1.4 Oscilloscope measuring the output signal from the microphone at 1kHz testing tone

To test my circuit, I used a 1kHz testing frequency from my phone to measure the peak voltages to see how large an output voltage from the microphone was. As can be seen in figure 1.4, the voltage peaks that were coming from the 1kHz testing tone was around 5 millivolts: way too low for what is needed for our applause meter.

The way we were able to do it was through an inverting amplifier. An inverting amplifier can boost our microphone signal through a combination of resistors. We use an inverting amplifier instead of a non-inverting amplifier for a variety of reasons. It offers more precise gain control and better signal processing capability [3]. Our amplification G is defined as $\frac{R_2}{R_1}$. In this case for the applause meter, seeing that the input voltage from the microphone was around 5 mV, it would be a safe measurement to have a gain of around 1000. This leads to $R_1 = 1\text{ k}\Omega$ and $R_2 = 1\text{ M}\Omega$.

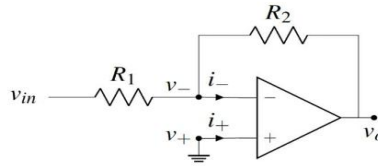


Figure 1.5 Inverting Amplifier [4]

Module 2:

The timed-reaction submodule is designed to provide a controlled delay for the output of the applause meter circuit. Its primary purpose is to maintain the output LED indicators for a predetermined period of time, ensuring that the physical photo-signal remains stable even after the clapping sound dissipates. Without such a feature, the collected peak signal from the microphone would rapidly decay, causing the LED indication to flicker and turn off instantly. To account for such an unwanted result, a capacitor-based timed reaction circuit is implemented along with a Schmitt-Trigger inverter and n-channel MOSFET switches.

The complete schematic includes 10 μ F capacitors, a CD40106B Schmitt-Trigger inverter, FQP30N06L nMOS transistors, 1k Ω pull-down resistors, and 100k Ω pull-up resistors. The final circuit schematic is shown in Fig. 2.1.

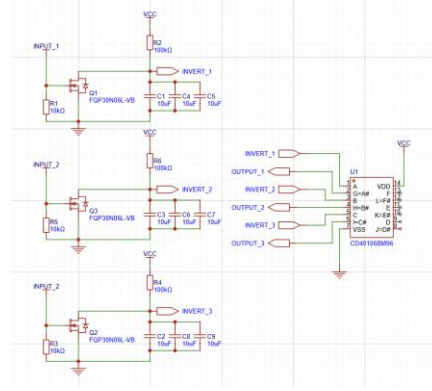


Fig. 2.1. Timed reaction schematic for LED output stabilization.

In the default state with the absence of clapping input, the pull-down resistors R_1 , R_2 , and R_3 maintain the MOSFET gates at low logic voltage, forcing each FQP30N06L nMOS into the cutoff region and impeding drain-to-source current. Simultaneously, the pull-up resistors connected to each capacitor charge the nodes to the supply voltage value. Consequently, the inputs of the CD40106B Schmitt-Trigger inverter read high logic voltage, producing a low voltage output and preventing the LEDs from activating [5], [6].

With a detection of a clapping input from the microphone, the input transitions to a high voltage, turning on the nMOS switches. The MOSFET gate voltage exceeds the threshold voltage specified in the FQP30N06L datasheet with $V_{GS(th)max} = 2.5 V$ [5], driving the device into the active region. The nMOS switches create a nearly short-circuit path for the capacitors to discharge, resulting in a low voltage level at the inverter input. The CD40106B outputs a high voltage, energizing the LEDs.

The discharge time constant is determined by the static drain-source on-resistance of the FQP30N06L, $R_{DS(on)} = 0.027 \Omega$ under $V_{GS} = 10 V$ [5], which is close to our actual supplied voltage, $V_{CC} = 9 V$. Therefore, the numerical value of the discharge time constant is expressed as

$$\tau_{dis} = R_{DS(on)}C = 0.027 \cdot 10 \mu = 0.27 \mu s$$

This extremely small time constant ($\tau_{dis} < 0.5 \mu s$) has a nearly negligible effect on the behavior of the circuit and is difficult to measure using standard oscilloscopes. Hence, it is excluded from calculations.

Following the discharge, the clapping peak voltage decays immediately back to its default low state. As a result, the MOSFET turns off, and the capacitor starts to recharge itself through the $100 k\Omega$ pull-up resistor. The theoretical charging time constant, τ_{char} , is expressed as

$$\tau_{char} = R_{up}C = 100 k \cdot 10 \mu = 1 s$$

The lab-recorded data yield a time constant of approximately 1.2416 s, which aligns with the theoretical calculation, given the multiple sources of error from resistors, capacitors, and the power supply.

Hence, to achieve the design goal specified by the rubric, a total of nine capacitors are used to achieve an output of at least three seconds at each channel.

To further validate the direct proportionality behavior of τ_{char} and C , a $100 \mu F$ capacitor was substituted for the previously used $10 \mu F$ capacitor. The newly measured charging duration increased to 9.6486 s, further reinforcing our equation for τ_{char} , with capacitance used directly proportional to the charging time constant value, $C \propto \tau_{char}$, with an approximate factor of 10.

To verify the RC behavior, the INPUT_1, INVERT_1, and OUTPUT_1 voltages were measured with the lab oscilloscope, as shown in Figure 2.2.

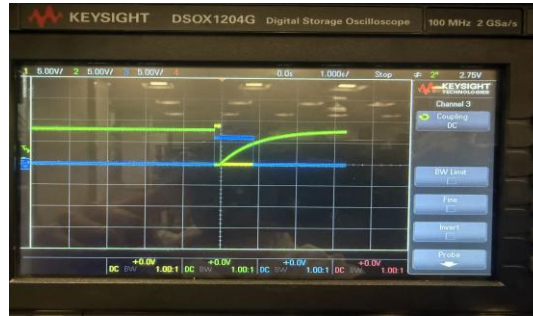


Fig. 2.2. Oscilloscope display of INPUT_1, INVERT_1, and OUTPUT_1 voltages.

The yellow trace represents the nMOS gate input from INPUT_1, while the green trace represents the capacitor voltage, and the blue trace corresponds to the OUTPUT_1 voltage.

When the input signal goes high, the nMOS turns on, creating a low-impedance path for the capacitor to drain to ground. This causes the capacitor's voltage to drop almost instantly. Because $R_{DS(on)} \ll R_{up}$ and $\tau_{dis} \ll \tau_{char}$, the voltage is displayed as a near-vertical drop on the oscilloscope display.

When the input signal goes low, the nMOS turns off, closing the low-impedance path and leaving the capacitor to charge through the $100\text{ k}\Omega$ pull-up resistor from V_{CC} . This creates an exponential rise in the capacitor's voltage. During the period of rise, the INVERT_1 pin would read a voltage lower than the CD40106B's positive trigger threshold voltage, $V_p = 5.9\text{ V}$ when $V_{DD} = 10\text{ V}$ [6]. For such a period of time, the CD40106B would read INVERT_1 as low voltage, thus outputting a high voltage at OUTPUT_1 for a set period of time, denoted by the blue trace. Such an output drives and holds the LED illumination for the final physical indication of the project.

Module 3:

The comparator stage converts the analog microphone signal into a clean digital pulse that can reliably be used by the timed reaction and LED-display circuitry. Conceptually, the entire comparator block can be seen as a one-bit quantizer with a tunable threshold. The LM311 itself is modeled as an extremely high gain differential stage followed by an open collector output transistor. The two input pins, labeled non-inverting and inverting, IN+ and IN- respectively can be seen in a pin diagram of the LM311.

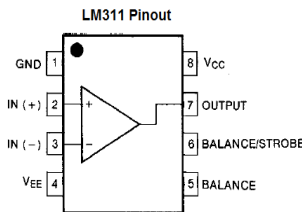


Fig 3.1. Pinout of LM311. Non-inverting(IN+) and inverting(IN-) can be seen to the left [7]

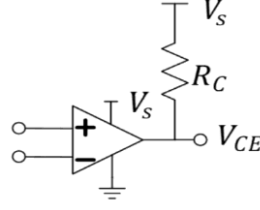


Fig 3.2. Circuit Schematic of our LM311[8]

As seen in figure 3.2, the non-inverting is tied to the amplified output. v_{mic} , and the inverting input is tied to a reference level, V_{ref} . In the ideal comparator model, the behavior can be written in logic terms as

$$v_{out}(t) = \begin{cases} 0V & v_{mic}(t) < V_{ref} \\ V_{cc} & v_{mic}(t) > V_{ref} \end{cases}$$

In our case, $V_{cc} = 9V$. Whenever V_{mic} is below V_{ref} , the LM311's internal transistor turns on and pulls its output pin down near ground. Whenever V_{mic} is above V_{ref} , that transistor turns off and an external resistance pulls the output pin up to that positive supply. Due to the high open loop gain of the LM311, the transition around the crossing is very steep and is reasonable to treat the switching as "instant" at the moment $V_{mic}(t) = V_{ref}$.

The reference voltage V_{ref} is created with a three terminal potentiometer used as a voltage divider between the +9V and ground. Each potentiometer has a distinct threshold that aligns with three different levels of sound. When choosing thresholds, we first found the minimum threshold all potentiometers had to be greater by checking V_{mic} with no artificial noise and only background, resulting in a peak of around 0.7V. Then we clapped at three distinct levels, measuring V_{mic} for each. After performing multiple trials, we concluded final values of 1.8, 3.6, and 6.2V.



Fig 3.3. Oscilloscope display of the potentiometer thresholds

Since the LM311 has an open collector output, it cannot drive the output high by itself. Instead, a resistor (R_C in figure 3.2) is connected between the output pin and V_{CC} . After testing, we found a R_C value of 100 k Ω worked best to minimize current draw while still getting edges that are "instant" since the node it drives is very high impedance and only has a small parasitic capacitance. With a 9V supply, a typical saturation voltage of 0.23V, a 100 k Ω pull-up limits the sink current in the LOW state to under 0.1 mA as seen below, which keeps power dissipation in the comparator small and avoids unnecessarily loading the chip [7].

$$I_{C,ON} = \frac{V_{CC} - V_{CE(sat)}}{R_C} = \frac{9 - 0.23}{100,000} = 0.08 \text{ mA}$$

When designing the circuit, we found initially the reaction timer was triggering randomly, however with further analysis with the oscilloscope we saw we underestimated the amplifier output and the background noise were

repeatedly grazing the reference level. We fixed this raising the potentiometer threshold so the background noise properly sat below threshold.

In the context of the whole circuit the comparator converts the continuous time, continuous amplitude signal from the analog front end in a binary ON/Off and determines how many LEDs turn on. The LM311 input presents a high impedance to the amplifier, so it does not disturb the analog gain or bandwidth, but its output behaves as a binary node. Since the comparator is unable to hold voltage, the output of the comparator feeds into the input of the timed reaction to ensure the LEDs stay on for the desired amount of time.

We found in some designs, engineers will add positive feedback to the comparator to create hysteresis (A Schmitt trigger), which gives one threshold for turning the output on and a slightly different one for turning it off again. That makes the comparator's output extremely stable when the input hovers near the boundary because the output will not chatter from noise. However, in our project, V_{ref} is high enough above the background noise that the input is either clearly below or clearly above it during claps. In later designs, maybe with more advanced noise detection, such implantation may be useful [9].

Module 4:

This module is on the peak detector. The purpose of a peak detector is to output the max or “peak” voltage from a time-varying voltage signal input. In the case of the applause meter, the original design was intended to have the input of the peak detector to be from the output of the microphone/amplifier component. This was a time-varying voltage signal that represented the signal from the clapping noise. The peak detector would then output the peak voltage from this signal into the timed reaction, where that peak voltage would be sustained. However, in later designs the peak detector was removed from the complete circuit design.

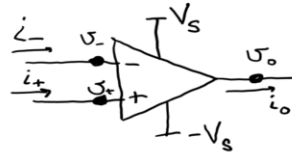


Figure 4.1 LM358 Op Amp Schematic Diagram [10]

Powering the Op Amp with +9 volts in the V+ pin and -9 volts in the GND pin allows the Op Amp to account for large positive and negative voltages. For this schematic $V_s = 9V$, $-V_s = -9V$, and V_+ is the time varying voltage input. We can assume that V_- and V_+ are equal when applying nodal analysis to the circuit [10].

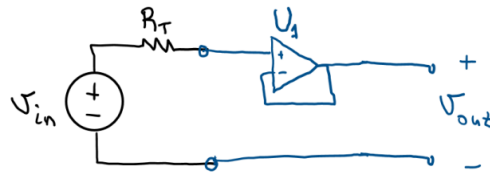


Figure 4.2 Voltage Follower Schematic with Op Amp [10]

Using this relationship, a voltage follower can be built by using V_- as a voltage output, meaning the voltage output would simply be equal to whatever voltage input is applied.

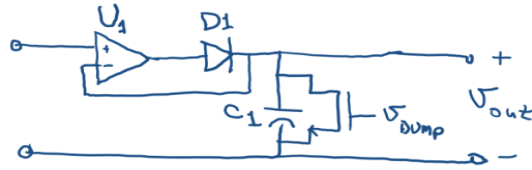
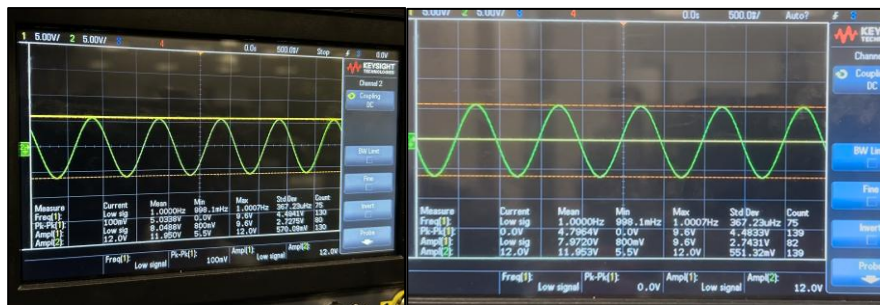


Figure 4.3 Peak Detector Schematic with NMOS [10]

The voltage follower is turned into a peak detector by adding a polarized capacitor that is charged through the V- node. This means that the voltage stored in the capacitor is equal to whatever the peak of the voltage input V+ is. A diode is added to prevent the capacitor from discharging to the Op Amp's output node. This means that the voltage output, which is equal to the capacitor's voltage, is equal to the peak of the voltage input, therefore creating a peak detector. The transistor is added to allow for the capacitor to drain, allowing for testing the peak detector's function. This design was built and demonstrated the expected behavior of the peak detector.



(a)

(b)

Figure 4.4 Oscilloscope Measurements of Peak Detector

Figure 4.4a displays the peak detector's output (in yellow) without the NMOS active. Figure 4.4b displays the peak detector's output with the NMOS active. The green line reading represents the voltage input into the peak detector.

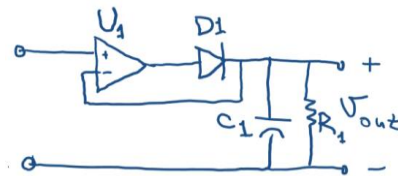


Figure 4.5 Peak Detector Schematic with Resistor [10]

In a later design, we also attempted using an alternative peak detector design where a resistor is placed in parallel with the capacitor. This design allows for a controlled path for the capacitor to discharge in a known amount of time. Which could be used in a circuit without a timed reaction component, where the output of the peak detector would go to the comparator. But the group collectively decided to work towards a circuit design that did not utilize a peak detector and instead used the timed reaction.

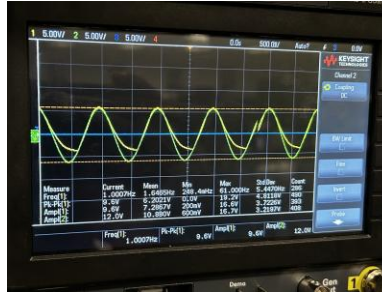


Figure 4.6 Oscilloscope measurements from *Figure 4.5 Peak Detector Design*

Figure 4.6 displays the intended output of this alternative peak detector design. In this case, the voltage output slowly begins to drain after the peak until it then detects another peak greater than its current voltage.

Design Choices

Initially our applause meter had the circuit in the following order:

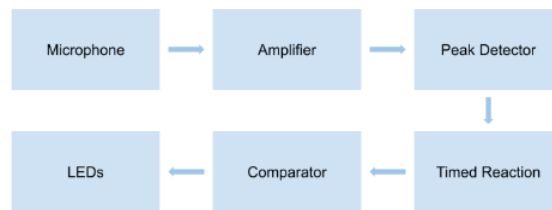


Figure 5.1 *Initial circuit design of our Applause Meter*

We initially thought that we needed the timed reaction to hold the voltage coming from the peak detector. So, when the peak detector had some output voltage V , the timed reaction would hold that voltage V for ideally 3 seconds which would be the input of the comparator and turn on specific LEDs depending on the voltage V .

However, as we continued testing and debugging the circuit, we realized that the peak detector was redundant in our applause meter as we just needed to only hold the value the comparator outputs. This means that we needed to have the comparator output be before the timed reaction so it can hold the value of the comparators for a set amount of time. The following was our final schematic design of what our applause meter would be like.

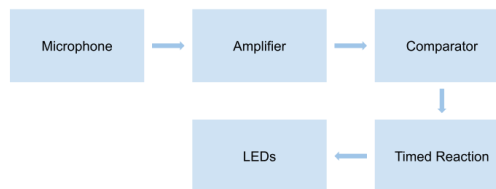


Figure 5.2 *Final circuit design of our Applause Meter*

Conclusions and Future Work

We successfully integrated all modules except the peak detector into a circuit that converts claps into an LED output. The condenser microphone and inverting op-amp provided a measurable voltage gain; the comparator turns that into a binary output and determines the strength of the signal; the CD40106 based time reaction network and nMOS switches power the LED and hold the voltage. While the peak detector behaves as expected in isolation, it was found that comparator and timed reactions met the project requirements more simply. We hope to apply the concepts learned in ECE 110 and the laboratory tools, like oscilloscopes and function generators, to test, analyze, and debug circuits in our journey as Electrical Engineers.

References

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